

Statistics for Artificial Intelligence

Appunti

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Chapter 1

Principal Component Analysis (PCA)

1.1 Empirical Covariance Matrices

Before deriving PCA, it is useful to make explicit the role of the *empirical covariance matrix*: it is the object PCA works on directly.

Setting. We have n observations $x_1, \dots, x_n \in \mathbb{R}^p$, i.e. n individuals on which we measure p variables. We write x_{ij} for the j -th variable of the i -th observation.

Definition 1.1.1: Empirical mean

The *empirical mean* of the dataset is the vector

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = \begin{bmatrix} \bar{x}_1 \\ \bar{x}_2 \\ \vdots \\ \bar{x}_p \end{bmatrix} \in \mathbb{R}^p, \quad \bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}.$$

Each component $\bar{x}_j$ is simply the mean of the j -th variable.

Definition 1.1.2: Empirical covariance matrix

The *empirical covariance matrix* is

$$S = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^\top \in \mathbb{R}^{p \times p}.$$

Entry by entry:

$$S_{jk} = \frac{1}{n} \sum_{i=1}^n (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k).$$

Note:

Each term $(x_i - \bar{x})(x_i - \bar{x})^\top$ is a *column* $\times$ *row* product ($p \times 1$ times $1 \times p$), hence a $p \times p$ matrix (of rank 1). S is the average of these n matrices.

Note:

n or $n-1$? Here we divide by n : this is the empirical-distribution convention (it coincides with the maximum-likelihood estimator in the Gaussian case). The usual *unbiased* estimator divides by $n-1$ instead:

$$S_{\text{unb}} = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^\top = \frac{n}{n-1} S.$$

For PCA **nothing changes**: the two matrices differ by a positive constant, so they have the same eigenvectors (the principal directions) and the same explained-variance ratios; only the numerical eigenvalues change, all multiplied by $\frac{n}{n-1}$.

1.1.1 Matrix form

Definition 1.1.3: Data matrix

The *data matrix* $X \in \mathbb{R}^{n \times p}$ contains the observations **by row**:

$$X = \begin{bmatrix} x_1^\top \\ \vdots \\ x_n^\top \end{bmatrix} \quad (\text{rows} = \text{observations, columns} = \text{variables}).$$

The data are *centered* if $\bar{x} = 0_p$, i.e. every column of X has zero mean.

Theorem 1.1.1

If the data are centered ($\bar{x} = 0_p$), then

$$S = \frac{1}{n} X^\top X.$$

In general, letting $X_c = X - \mathbf{1}_n \bar{x}^\top$ be the centered matrix ($\bar{x}^\top$ is subtracted from every row), we always have $S = \frac{1}{n} X_c^\top X_c$.

Proof: The product $X^\top X$ can be written as a sum of column $\times$ row products: the i -th column of $X^\top$ is x_i and the i -th row of X is $x_i^\top$, so

$$X^\top X = \sum_{i=1}^n x_i x_i^\top.$$

If $\bar{x} = 0_p$, then $x_i - \bar{x} = x_i$ and the definition of S becomes $S = \frac{1}{n} \sum_i x_i x_i^\top = \frac{1}{n} X^\top X$. In the general case, apply the same computation to X_c , whose rows are $(x_i - \bar{x})^\top$. $\square$

Note:

This is why in practice **data are always centered** before PCA: from then on the covariance is simply $\frac{1}{n} X^\top X$.

1.1.2 How to read S

The covariance matrix summarizes how the variables vary, individually and jointly:

- **diagonal:** $S_{jj} = \frac{1}{n} \sum_i (x_{ij} - \bar{x}_j)^2$ is the *variance* of variable j ;
- **off-diagonal:** S_{jk} is the *covariance* between variables j and k ;
 - $S_{jk} > 0$: when j is above its mean, k tends to be above its mean too;
 - $S_{jk} < 0$: when one goes up, the other tends to go down;
 - $S_{jk} \approx 0$: no *linear* association.

Example 1.1.1

If $p = 3$ (variables: height, weight, age):

$$S = \begin{bmatrix} \text{Var}(\text{height}) & \text{Cov}(\text{height, weight}) & \text{Cov}(\text{height, age}) \\ \text{Cov}(\text{weight, height}) & \text{Var}(\text{weight}) & \text{Cov}(\text{weight, age}) \\ \text{Cov}(\text{age, height}) & \text{Cov}(\text{age, weight}) & \text{Var}(\text{age}) \end{bmatrix}.$$

Note:

The value of S_{jk} depends on the units of measurement: a “large” covariance must be judged *relative to the scales* of the variables. To compare, one uses the correlation

$$r_{jk} = \frac{S_{jk}}{\sqrt{S_{jj} S_{kk}}} \in [-1, 1].$$

1.1.3 Basic properties

Proposition 1.1.1 Symmetry

$$S^\top = S.$$

Proof: Each term is symmetric: $((x_i - \bar{x})(x_i - \bar{x})^\top)^\top = (x_i - \bar{x})(x_i - \bar{x})^\top$, and a sum of symmetric matrices is symmetric. $\square$

Proposition 1.1.2 Positive semidefiniteness

For every vector $a \in \mathbb{R}^p$,

$$a^\top S a \geq 0.$$

More precisely, $a^\top S a$ is the empirical variance of the data projected along a , i.e. of the numbers $a^\top x_1, \dots, a^\top x_n$.

Proof:

$$a^\top S a = \frac{1}{n} \sum_{i=1}^n a^\top (x_i - \bar{x})(x_i - \bar{x})^\top a = \frac{1}{n} \sum_{i=1}^n (a^\top x_i - a^\top \bar{x})^2 \geq 0,$$

since $a^\top (x_i - \bar{x})$ is a scalar, so it equals its transpose $(x_i - \bar{x})^\top a$. The last expression is exactly the empirical variance of the values $a^\top x_i$, whose mean is $a^\top \bar{x}$. $\square$

Corollary 1.1.1 Nonnegative eigenvalues

S has p real eigenvalues $\lambda_1 \geq \dots \geq \lambda_p \geq 0$ and an orthonormal basis of eigenvectors.

Proof: Since S is symmetric, the spectral theorem gives real eigenvalues and an orthonormal basis of eigenvectors. If $Sv = \lambda v$ with $\|v\| = 1$, then by positive semidefiniteness

$$\lambda = \lambda v^\top v = v^\top S v \geq 0. \quad \square$$

Note:

This is why positive semidefiniteness is crucial: eigenvalues of S are *variances* (along the eigenvector directions), so they cannot be negative. It is also the bridge to PCA: finding the direction a (with $\|a\| = 1$) along which the data vary the most means maximizing $a^\top S a$.

Proposition 1.1.3 Rank bound

If $X \in \mathbb{R}^{n \times p}$, then

$$\text{rank}(X^\top X) = \text{rank}(X) \leq \min(p, n).$$

For centered data, $S = \frac{1}{n} X_c^\top X_c$ satisfies the sharper bound

$$\text{rank}(S) \leq \min(p, n - 1).$$

Proof: $\text{rank}(X^\top X) = \text{rank}(X)$. We show $\ker(X^\top X) = \ker(X)$. If $Xv = 0$ then clearly $X^\top Xv = 0$. Conversely, if $X^\top Xv = 0$ then

$$0 = v^\top X^\top X v = \|Xv\|^2 \implies Xv = 0.$$

Both matrices have p columns, so by rank-nullity they have the same rank. And $\text{rank}(X) \leq \min(p, n)$ because X has n rows and p columns.

Centered case. Centering imposes one linear constraint on the rows: they sum to zero,

$$\sum_{i=1}^n (x_i - \bar{x}) = n\bar{x} - n\bar{x} = 0, \quad \text{i.e.} \quad \mathbf{1}_n^\top X_c = 0.$$

So the n rows of X_c are linearly dependent and span a space of dimension at most $n - 1$. Hence $\text{rank}(S) = \text{rank}(X_c) \leq \min(p, n - 1)$ (the factor $\frac{1}{n}$ does not change the rank). $\square$

Note:

High-dimensional setting. When $n \ll p$ (few observations, many variables), the rank bound gives $\text{rank}(S) \leq n - 1 < p$: S is **low rank**, hence singular, with at least $p - n + 1$ zero eigenvalues. This situation is called a *high-dimensional setting* and requires special treatment.

Example 1.1.2

With $n = 2$ observations in $\mathbb{R}^3$, after centering $x_2 - \bar{x} = -(x_1 - \bar{x})$: all the data lie on a single line through $\bar{x}$, so $\text{rank}(S) \leq 1$ even though S is 3×3 .

1.2 Maximum Variance Principle

The main objective of PCA is to find a **low-dimensional subspace** such that the **projected data have maximum variance**. The idea:

- if the projected points still *spread out* substantially, the projection retains information;
- if the projected points *collapse* into a small region, much structure has been removed.

From now on the data are **centered** ($\bar{x} = 0_p$) and $S = \frac{1}{n} X^\top X$.

Note:

Notation clash. The course uses p both for the number of variables and for a direction vector. To keep them apart, here the direction is written in bold: $\mathbf{p} \in \mathbb{R}^p$.

1.2.1 One direction

Definition 1.2.1: Latent score

Given a direction $\mathbf{p} \in \mathbb{R}^p$, the *latent score* of observation i is

$$z_i = \mathbf{p}^\top x_i \in \mathbb{R},$$

i.e. the coordinate of x_i along $\mathbf{p}$ (its projection, when $\|\mathbf{p}\| = 1$).

By the positive semidefiniteness proposition (the data are centered, so the scores have mean 0), the variance of the latent scores is

$$\text{var}(z) = \frac{1}{n} \sum_{i=1}^n z_i^2 = \mathbf{p}^\top S \mathbf{p}.$$

Note:

Why the constraint $\mathbf{p}^\top \mathbf{p} = 1$. Without it the problem has no finite maximum: for any scalar $c \in \mathbb{R}$,

$$\text{var}(cz) = (c\mathbf{p})^\top S(c\mathbf{p}) = c^2 \mathbf{p}^\top S \mathbf{p},$$

which, whenever $\mathbf{p}^\top S \mathbf{p} > 0$, becomes arbitrarily large by taking $|c|$ large. The constraint **fixes the scale**, so that we compare only *directions*, not lengths.

Suggested exercises: 23 and 24.

Definition 1.2.2: PCA optimization problem (first direction)

$$\max_{\mathbf{p} \in \mathbb{R}^p} \mathbf{p}^\top S \mathbf{p} \quad \text{subject to} \quad \mathbf{p}^\top \mathbf{p} = 1.$$

Theorem 1.2.1

Every stationary point of the problem satisfies the eigenvalue equation

$$S\mathbf{p} = \lambda\mathbf{p},$$

and at such a point the objective equals $\mathbf{p}^\top S \mathbf{p} = \lambda$. Therefore:

- the **principal directions** are (unit) **eigenvectors** of S ;
- the **variance along each principal direction** is the corresponding **eigenvalue**;
- the maximum is λ_1 , the largest eigenvalue, attained at its eigenvector $\mathbf{p}_1$.

Proof: Lagrange multiplier. Consider

$$\mathcal{L}(\mathbf{p}, \lambda) = \mathbf{p}^\top S \mathbf{p} - \lambda (\mathbf{p}^\top \mathbf{p} - 1).$$

Since S is symmetric, $\nabla_{\mathbf{p}} \mathbf{p}^\top S \mathbf{p} = 2S\mathbf{p}$ and $\nabla_{\mathbf{p}} \mathbf{p}^\top \mathbf{p} = 2\mathbf{p}$, so

$$\nabla_{\mathbf{p}} \mathcal{L} = 2S\mathbf{p} - 2\lambda\mathbf{p} = 0 \iff S\mathbf{p} = \lambda\mathbf{p}.$$

Multiplying on the left by $\mathbf{p}^\top$ and using $\mathbf{p}^\top \mathbf{p} = 1$: $\mathbf{p}^\top S \mathbf{p} = \lambda$. Among the stationary points, the best one is the eigenvector with the largest eigenvalue.

Direct check that λ_1 is the maximum. Write $S = P\Lambda P^\top$ (see below) and $\mathbf{y} = P^\top \mathbf{p}$; since P is orthogonal, $\|\mathbf{y}\| = \|\mathbf{p}\| = 1$. Then

$$\mathbf{p}^\top S \mathbf{p} = \mathbf{y}^\top \Lambda \mathbf{y} = \sum_{j=1}^p \lambda_j y_j^2 \leq \lambda_1 \sum_{j=1}^p y_j^2 = \lambda_1,$$

with equality for $\mathbf{y} = \mathbf{e}_1$, i.e. $\mathbf{p} = \mathbf{p}_1$. ☺

Suggested exercise: 25.

1.2.2 Spectral decomposition and q directions

Theorem 1.2.2

(Spectral decomposition.) Since S is symmetric,

$$S = P\Lambda P^\top,$$

where

- $P = [\mathbf{p}_1 \mid \cdots \mid \mathbf{p}_p] \in \mathbb{R}^{p \times p}$ is orthogonal ($P^\top P = I$), its columns are the orthonormal eigenvectors $\mathbf{p}_j$;
- $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_p)$ contains the eigenvalues, ordered as

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_p \geq 0$$

(nonnegative by the corollary on eigenvalues).

To project onto a q -dimensional subspace ($q < p$) we take an orthonormal basis $W = [\mathbf{w}_1 \mid \cdots \mid \mathbf{w}_q] \in \mathbb{R}^{p \times q}$, $W^\top W = I_q$. The variance retained is the sum of the variances of the q scores:

$$\sum_{k=1}^q \mathbf{w}_k^\top S \mathbf{w}_k = \text{tr}(W^\top S W).$$

Theorem 1.2.3

(Best q -dimensional subspace.) For every $W \in \mathbb{R}^{p \times q}$ with $W^\top W = I_q$,

$$\text{tr}(W^\top S W) \leq \lambda_1 + \cdots + \lambda_q,$$

with equality for $W = [\mathbf{p}_1 \mid \cdots \mid \mathbf{p}_q]$. Hence the best q -dimensional PCA subspace is **spanned by the eigenvectors of the q largest eigenvalues**.

Proof: Let $Y = P^\top W$ (still with orthonormal columns). Then

$$\text{tr}(W^\top S W) = \text{tr}(Y^\top \Lambda Y) = \sum_{j=1}^p \lambda_j h_j, \quad h_j = \sum_{k=1}^q Y_{jk}^2.$$

The weights satisfy $0 \leq h_j \leq 1$ (h_j is the squared norm of the projection of e_j onto the column space of Y) and $\sum_j h_j = \text{tr}(Y^\top Y) = q$. With q units of weight, each λ_j receiving at most 1, the sum $\sum_j \lambda_j h_j$ is largest by putting weight 1 on the q largest eigenvalues: $\sum_j \lambda_j h_j \leq \lambda_1 + \cdots + \lambda_q$. For $W = [\mathbf{p}_1 \mid \cdots \mid \mathbf{p}_q]$ we get $h_1 = \cdots = h_q = 1$, so equality holds. $\square$

Definition 1.2.3: Captured and lost variance

The variance *captured* by the first q principal components is

$$V_q = \sum_{j=1}^q \lambda_j,$$

and the variance *lost* is

$$J_{p-q} = \sum_{j=q+1}^p \lambda_j.$$

Note:

$V_q + J_{p-q} = \sum_{j=1}^p \lambda_j = \text{tr}(S) = \sum_j S_{jj}$, the *total variance* (sum of the variances of all variables). So the fraction of **explained variance** is

$$\frac{V_q}{\text{tr}(S)} = \frac{\lambda_1 + \cdots + \lambda_q}{\lambda_1 + \cdots + \lambda_p}.$$

This ratio is unchanged if S is rescaled (e.g. n vs $n-1$).

Example 1.2.1

Suppose $p = 3$ and S has eigenvalues $\lambda_1 = 5$, $\lambda_2 = 2$, $\lambda_3 = 1$, so $\text{tr}(S) = 8$.

- $q = 1$: $V_1 = 5$, $J_2 = 3$, explained variance $5/8 = 62.5\%$;
- $q = 2$: $V_2 = 7$, $J_1 = 1$, explained variance $7/8 = 87.5\%$.

Projecting onto the plane spanned by $\mathbf{p}_1, \mathbf{p}_2$ loses only 12.5% of the variance.

1.3 A Worked Example in R

We apply everything to a simulated dataset with $n = 150$ observations and $p = 2$ variables.

```
set.seed(123)
n <- 150
x1 <- rnorm(n, sd = 2)
x2 <- 0.6 * x1 + rnorm(n, sd = 0.5) # x2 depends linearly on x1, plus noise
X <- cbind(x1, x2)
```

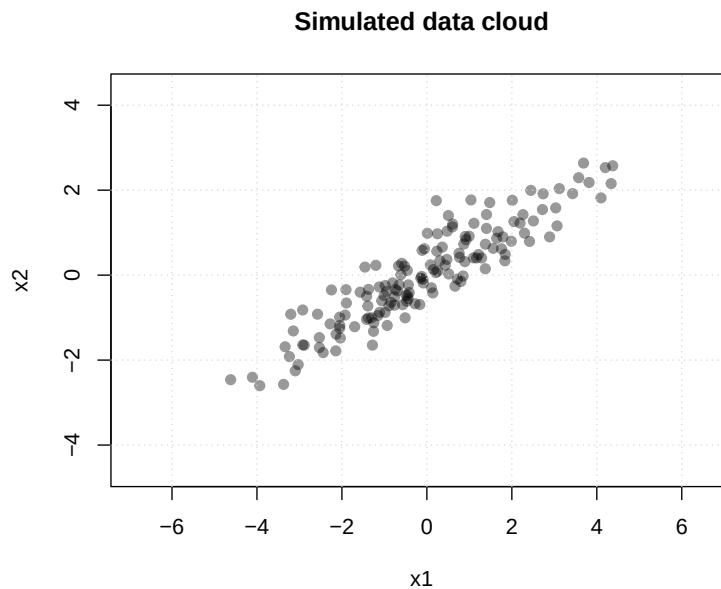


Figure 1.1: The simulated data cloud.

1.3.1 Interpretation of the scatter plot

This graph is the starting point. PCA is a **geometric** method, so we first look at the *shape* of the data cloud: if the cloud is **elongated**, most of the variation may lie close to one direction. PCA formalizes that intuition.

1.3.2 Covariance matrix and spectral decomposition

We center the data and compute $S = \frac{1}{n} X_c^T X_c$, then its eigen-decomposition:

```
X_centered <- scale(X, center = TRUE, scale = FALSE) # X_c: subtract column means
S <- crossprod(X_centered) / nrow(X_centered) # crossprod(A) = t(A) %*% A

S
## x1 x2
## x1 3.584051 2.004268
## x2 2.004268 1.334890

eig <- eigen(S)
eig$values
## [1] 4.757681 0.161260
eig$vectors
## [,1] [,2]
## [1,] -0.8629393 0.5053075
## [2,] -0.5053075 -0.8629393
```

Example 1.3.1

Reading the output with the theory:

- $S_{11} \approx 3.58$ and $S_{22} \approx 1.33$ are the variances; $S_{12} \approx 2.00 > 0$: x_1 and x_2 grow together (correlation $2.00/\sqrt{3.58 \cdot 1.33} \approx 0.92$);
- $\lambda_1 \approx 4.76$, $\lambda_2 \approx 0.16$: almost all variance is along $\mathbf{p}_1$;
- the columns of `eig$vector`s are $\mathbf{p}_1, \mathbf{p}_2$: unit length and orthogonal;
- check: $\lambda_1 + \lambda_2 = 4.918941 = S_{11} + S_{22} = \text{tr}(S)$.

Note:

Sign of eigenvectors. If $\mathbf{p}$ is a unit eigenvector, so is $-\mathbf{p}$. Here R returns $\mathbf{p}_1 = (-0.86, -0.51)^\top$, but $(0.86, 0.51)^\top$ is equally valid. So principal directions and scores are defined only *up to sign*.

Note:

n vs $n - 1$ again. Here S uses denominator n . By contrast, `prcomp()` reports component variances with denominator $n - 1$ (R's sample-covariance convention). This multiplies the eigenvalues by $n/(n - 1)$ but does **not** change the principal directions, the latent scores (up to sign), the reconstructions or the explained-variance ratios.

1.3.3 Principal directions

We draw each principal direction as an arrow:

```
mu <- colMeans(X)
P <- eig$vector
lambda <- eig$value

arrows_df <- data.frame(
  x0 = mu[1], y0 = mu[2],
  x1 = mu[1] + 2 * sqrt(lambda[1]) * P[1, 1],
  y1 = mu[2] + 2 * sqrt(lambda[1]) * P[2, 1],
  x2 = mu[1] + 2 * sqrt(lambda[2]) * P[1, 2],
  y2 = mu[2] + 2 * sqrt(lambda[2]) * P[2, 2]
)
## x0 y0 x1 y1 x2 y2
## x1 -0.04872574 0.0173874 -3.813231 -2.186977 0.3571088 -0.6756769
```

(then plotted with `ggplot2`: `geom_point` for the data and one `geom_segment` with an arrow for each direction.)

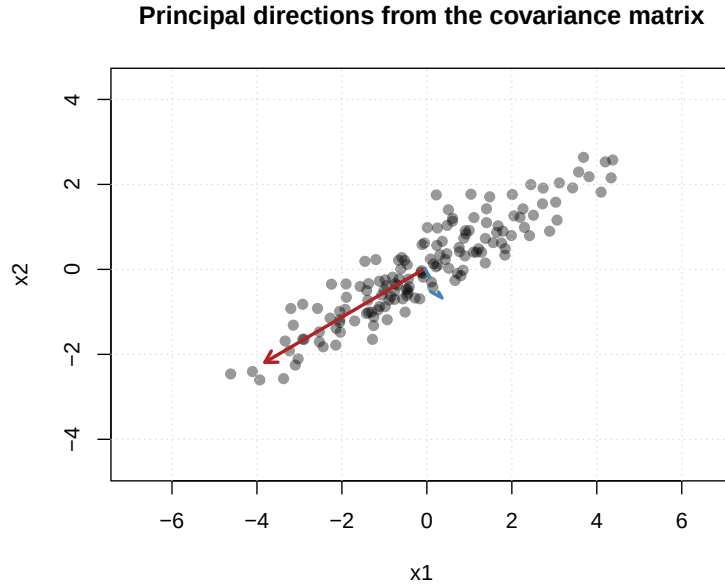


Figure 1.2: First principal direction (solid, red) and second (dashed, blue).

Interpretation

Each arrow is

$$\underbrace{\mu}_{\text{start}} + \underbrace{2\sqrt{\lambda_j}}_{\text{length}} \underbrace{p_j}_{\text{direction}}, \quad \text{i.e. in R: } \text{mu} + 2 * \text{sqrt}(\text{lambda}[j]) * \text{P[, } j].$$

- **start:** the empirical mean $\mu = \frac{1}{n} \sum_i x_i$, the center of the cloud; this is why both arrows start at $(\text{mu}[1], \text{mu}[2])$;
- **direction:** the eigenvector, $P[, 1]$ for the first principal direction and $P[, 2]$ for the second;
- **length:** λ_j is the *variance* along p_j , so $\sqrt{\lambda_j}$ is a *standard deviation*, which has the same units as the data. The factor 2 is only a visualization choice to make the arrows clearly visible.

1.3.4 Explained variance ratio

```
| prop_var <- eig$values / sum(eig$values)
```

Component	Eigenvalue	Explained variance ratio
1	4.7577	0.9672
2	0.1613	0.0328

In this example **one component already explains 96.7% of the variability**: with $q = 1$, $V_1 = \lambda_1$ and the lost variance is only $J_1 = \lambda_2$.

1.3.5 Projection and reconstruction

With one retained component, each observation is projected onto the line through μ spanned by the first principal direction.

Definition 1.3.1: Score and reconstruction ($q = 1$)

The *score* of observation i on the first component and its *reconstruction* are

$$z_i = \mathbf{p}_1^\top (x_i - \mu), \quad \hat{x}_i = \mu + z_i \mathbf{p}_1 = \mu + \mathbf{p}_1 \mathbf{p}_1^\top (x_i - \mu).$$

The *reconstruction error* is the vector $x_i - \hat{x}_i$.

We compute this with `prcomp()`, where `$x` are the scores z_i , `$rotation` the loadings (the matrix P) and `$center` the mean μ :

```
pca_fit <- prcomp(X, center = TRUE, scale. = FALSE)

scores1 <- pca_fit$x[, 1, drop = FALSE] # z_i (n x 1)
loadings1 <- pca_fit$rotation[, 1, drop = FALSE] # p_1 (2 x 1)

Xhat1 <- scores1 %*% t(loadings1) # z_i p_1^T (n x 2)
Xhat1 <- sweep(Xhat1, 2, pca_fit$center, "+") # add mu back to each row

head(data.frame(x1 = X[, 1], x2 = X[, 2],
                x1_hat = Xhat1[, 1], x2_hat = Xhat1[, 2]))
## x1 x2 x1_hat x2_hat
## 1 -1.1209513 -0.27870135 -0.97628325 -0.52575832
## 2 -0.4603550 0.10830813 -0.31560537 -0.13888812
## 3 3.1174166 2.03655127 3.18944333 1.91354762
## 4 0.1410168 -0.41957823 -0.09797001 -0.01144835
## 5 0.2585755 0.09541898 0.21413615 0.17131026
## 6 3.4301300 1.91788032 3.37056335 2.01960526
```

Projection onto the first principal component

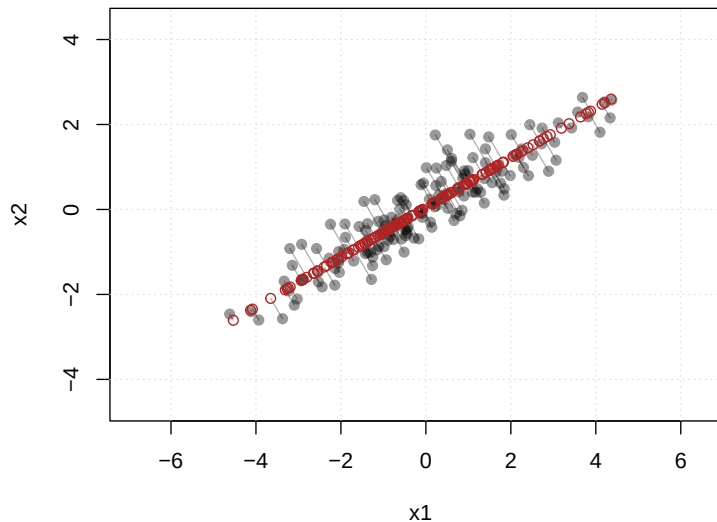


Figure 1.3: Original points (black), reconstructions $\hat{x}_i$ (red circles) and reconstruction errors (segments).

Interpretation

There are three geometric objects in this figure:

- the **original data points** x_i (x_1 , x_2);

- the **reconstructed points** $\hat{x}_i$ (`x1_hat`, `x2_hat`): the projections of the original observations onto the first principal component, so they all lie on one line;
- the **segments** between them, each representing the reconstruction error $x_i - \hat{x}_i$.

This is why the graph is useful: it shows visually what PCA does when only one component is kept.

Note:

The segments are **perpendicular** to the line: $x_i - \hat{x}_i$ is the component of $x_i - \mu$ along $\mathbf{p}_2$. Hence the average squared length of the segments is exactly the lost variance:

$$\frac{1}{n} \sum_{i=1}^n \|x_i - \hat{x}_i\|^2 = \lambda_2 = J_1 \approx 0.16.$$

Maximizing the variance kept is the same as minimizing the reconstruction error.